MINAR

Learn money by playing it · M1 Approach

Team 151

1 Problem and our understanding

Design a digital experience Grade 11–12 students **want** to use, on financial literacy, for underserved communities in Hyderabad. Start with ~1,500 students; design to scale.

The real barrier
Not information but relevance and trust. Lectures on "compounding" lose a 16-year-old. A safe place to choose, see the consequence and retry does not.

What we must cover
4 pillars: **Plan & budget · Save & manage · Understand investing · Stay safe**, judged on 7 measures: simple, relatable, persona-led, language-ready, game-based, intuitive, NGO-integrated.

Hard guardrail: we educate, we never recommend investments. Simulations are illustrative, with fictional companies and no real money, brands or personal data.

2 Idea in one line

Minar is a mobile-first game set in a painted Hyderabad bazaar. Walk to the Charminar's four shops; each opens a short mini-game. Finish a pillar and a minaret lamp lights up. It grades **judgement, not profit**, and builds a six-skill profile: planning, prioritising, saving, risk sense, patience, scam sense.

4	3	3	0	360px
pillar mini-games	personas	languages EN/HI/TE	sign-ins or PII	low-end phone ready

Pillar	Shop	Game	Core mechanic and what it teaches
1 Plan & budget	Bangle stall	Weekly Budget	Split a week's money across needs, wants, goals; surprises test the plan
2 Save & manage	Irani café	Two Bad Weeks	Run small income through emergencies: spend, save or borrow
3 Investing	Pearl shop	Deal Board	Monopoly-style board: ₹1,00,000 play money, 12 quarters, risk, time, spread
4 Stay safe	UPI kiosk	Spot the Scam	Triage fake offers, OTP and link messages; learn the red flags

3 Pipeline and architecture

Onboard persona + language › **Hub** walkable bazaar › **Mini-game** seeded engine › **Consequence** 3-line lesson › **Retry** rewind and replay › **Report** skills + NGO view

Engine and UI split
Each game is a pure TypeScript engine with a seeded RNG: same seed and choices give an identical run, so it is testable and fair to replay.

Static, zero backend
Vite + React + TypeScript + Tailwind; state in localStorage. Free to host, scales to any cohort, offline-friendly.

Persona-led
School, college or first-job persona sets starting money, goals, wording and difficulty.

Language-ready
All copy in locale files (English, Hindi, Telugu); plain words; icon plus label for every signal.

4 Flagship: Deal Board (Pillar 3)

- 16-tile ring, seeded dice, 12 turns (quarters), fictional companies.
- Each firm shows **4 lights**: Cash, Customers, Debt, CEO answer (green, amber, red, grey), always icon plus label.
- Research by asking the founder (flip 2 lights), then **bid or pass**.
- Real-life frictions: 1% fees, idle cash loses value, a loan, a market fall, a ₹25,000 emergency.

Skill graded (20 each)	Shown by
Research	Checks lights before bidding
Skepticism	Passes on red flags
Spread	Avoids one-sector bets
Calm and patience	No panic-sell in a fall
Money habits	Fees, loan, reserve

A stage is cleared at a **judgement score of 60/100**, never by profit. Every sim screen carries: *"Simulation with made-up numbers. Real life is different."*

5 Fit to the success measures

Measure	How Minar delivers it
Simple and intuitive	Walk, tap, play. No manual; first decision in under 30 seconds; joystick or tap-to-walk
Relatable	Hyderabad bazaar, ₹ amounts, chai, UPI, recharge scams
Persona-led	3 personas scale amounts, goals and scenarios
Language-ready	EN / HI / TE from one string table; native-speaker review
Game-based evaluation	Skill score per game, rewind and retry, end-of-pillar recap
NGO-integrated	Cohort code, PII-free CSV of completion and skill levels, facilitator dashboard

6 Planned testing methodology

Layer	What we check	How
Engine	Determinism, fees, light rules, scoring, emergency timing	Vitest unit tests on every commit
Balance	Skill beats luck; no dominant exploit	Bots over 1,000 seeds: equal-spread bot scores 35–55, research bot beats random
Learning	Concepts understood, not memorised	Pre/post scenario questions; skill gain on replay; lesson cards fire on triggers
UX and access	Works for beginners on cheap phones	Playwright end-to-end run; 360px to desktop; lite mode; peer walkthroughs
Language	Meaning and fit in HI and TE	Key-coverage script, text-fit screenshots, native review
Guardrails	No advice, no real brands, no PII	Content audit; banner test on every sim screen; storage audit

7 Success looks like

<30s	**5–8 min**	**↑ score**	**1,500+**
to first decision, no help	per pillar	on replay	students, no backend